Code No: 133BD **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, March - 2021
MATHEMATICS – IV
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MCT, ETM, MMT, AE, MIE, PTM, CEE, MSNT)
Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Find the analytic function whose real part is $e^x x\cos y \quad e^x y\sin y$.
b) Show that if u is harmonic and v is a harmonic conjugate of u, then u is a harmonic conjugate of v. [7+8]

2.a) Show that the function $x^2 + iy^3$ is not analytic anywhere. Reconcile this with the fact that the Cauchy–Riemann equations are satisfied at $x = 0, y = 0$.
b) Evaluate $\int_{(0,1)}^{(2,5)}(3x + y)dx + (2y - x)dy$ along:
i) the curve $y = x^2 + 1$,
ii) the straight line joining (0, 1) and (2, 5),
iii) the straight lines from (0, 1) to (0, 5). [7+8]

3.a) Integrate counterclockwise around the unit circle. Indicate whether Cauchy's integral theorem applies. Show the details.
$f(z) = \overline{z^3}$
$f(z) = tan(1/4z)$
b) Integrate by Cauchy's integral formula $z^2/(z^2 - 1)$ counter clockwise around the circles $|z + 1| = 1$ and $|z + 1 - i| = \pi/2$. [8+7]

4.a) Find the Taylor series with center z_0 and find its radius of convergence.
$1/(1 - z), \quad z_0 = i$.
b) Find the Laurent series that converges for $0 < |z - \ z_0| < R$ and determine the precise region of convergence. Show the details $\frac{e^z}{(z-1)^2}$, $z_0 = 1$. [8+7]

5.a) Find all the singularities in the finite plane and the corresponding residues $\frac{sin2z}{z^6}$
b) Evaluate counterclockwise $\oint_c e^{1/z}dz\, C$: the unit circle. [8+7]

6. Evaluate the following:
a) $\int_0^{2\pi} \frac{sin^2\theta}{(5-4cos\theta)}\, d\theta$
b) $\int_0^{\infty} \frac{dx}{(1+x^2)^3}$ [8+7]

Used Papers 08-03-21AM

7.a) Is the given function even or odd or neither even nor odd? Find its Fourier series.
$f(x) = x^2, (-1 < x < 1), p = 2$
And hence show that $(1/6)\pi^2 = 1 + 1/4 + 1/9 + 1/16$ ....

b) Find the cosine transform of $\widehat{f_c}(w)$ of
$f(x) = 1$ if $0 < x < 1$
$f(x) = -1$ if $1 < x < 2$
$f(x) = 0 \quad if \quad x > 2$ [8+7]

8. Find the deflection $u(x,t)$ for the string of length L and $c^2 = 1$ when the initial velocity is zero and the initial deflection with small k (say, 0.01) is $k(sin\pi x - (1/2)sin2\pi x)$. [15]

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